NullState Phase 1 — Implementation-Grade Protocol for the Two Hardest Pieces: Real scArches Surgery on Recovered Counts, and contrastiveVI Dish-Stripping with a Maturation-Matched Control

TL;DR

- **Hard Piece 1 is solvable and is the gating dependency:** recover genuine per-study raw integer UMIs (the public HNOCA object stores `lognorm`/`scvi_normalized`/embedding layers, not documented raw counts), then run *real* scArches surgery with `weight_decay=0.0`, `gene_likelihood='nb'` (not `zinb`), partial freezing and ~150 capped epochs with ELBO early-stopping. The central risk is **over-correction** that drags genuinely off-manifold off-target cells onto the reference manifold and destroys the signal — detect it with paired pre/post latent-mixing diagnostics (batch-mixing iLISI up on on-target cells, but cell-type cLISI and a mesenchyme marker-coherence check preserved).
- **Hard Piece 2 is the scientific crux and the deepest uncertainty:** contrastiveVI (background = primary on-target, target = organoid on-target) isolates the dish axis into a salient latent; prove disentanglement quantitatively (origin AUROC >0.85 on salient, ≈0.5 on shared; salient correlates with HALLMARK_GLYCOLYSIS/HYPOXIA/UNFOLDED_PROTEIN_RESPONSE, shared does not). The decisive **Maturation-Matched Control 3** uses CytoTRACE2 potency bins to build a maturation-matched primary baseline, then tests convergence with **sliced-Wasserstein distance in the dish-stripped shared latent**, with a permutation null shuffling labels *within* maturation bins. If matching collapses the signal to the self-distance floor, the result must be reframed as a *shared-immature/arrested state*, not an "attractor."
- **Both fit the USD 150–200 budget** on a single A100/H100 spot instance (≈$1–3/GPU-hr in 2026): surgery and contrastiveVI each train in a few GPU-hours; the dominant cost is wall-clock for CPU permutation/rarefaction. Every result must be wrapped in **leave-one-study-out** because the off-target mesenchyme is study-concentrated.

Key Findings

1. **The pilot's NaN divergence was a likelihood mis-specification, not a tuning failure.** scVI/scANVI's NB/ZINB observation model is defined only for non-negative integers; feeding log-normalized or length-normalized values makes the NB log-likelihood and its gradients unstable, producing NaN/inf that propagate through backprop. The fix is genuine counts plus a more robust likelihood (`nb`, or Poisson fallback), not lowering the learning rate alone.
2. **HNOCA was integrated with scPoli, and its public layers are not documented to contain raw integer counts.** Per He, Dony, Fleck et al. (Nature 635:690–698, 2024), the atlas comprises "1.77 million cells … from 36 datasets" (34 published, 2 unpublished), and the Methods state verbatim: *"We performed integration of the organoid datasets for HNOCA using the scPoli model from the scArches package … [batch covariate from] dataset identifier … bio_sample … tech_sample. This resulted in 396 individual batches."* The HNOCA-tools workflow references a `lognorm` layer and an `scvi_normalized` layer; raw counts must be recovered per-study from the original GEO/ArrayExpress deposits (accessions in Supplementary Table 1, curated through sfaira). This is the single most important — and least certain — operational step. GitHub
3. **Surgery over-correction is a real and under-appreciated failure mode** distinct from divergence. scArches (Lotfollahi, Naghipourfar, Luecken et al., Nat Biotechnol 40:121–130, 2022) is built to remove batch while *preserving* biological state "despite using four orders of magnitude fewer parameters than de novo integration," but with limited reference coverage of non-neural lineages it can absorb true off-target structure into the batch term. The diagnostic that distinguishes over- from under-correction is the *combination* of batch-mixing (iLISI/kBET up) with biological-conservation (cLISI, marker coherence) preserved.
4. **A single salient subspace probably under-models the dish signature** because glycolysis/hypoxia are pan-organoid while ER/UPR stress is lineage-modulated. Bhaduri et al. (Nature 578:142–148, 2020) report that "ER stress and glycolysis networks decrease over time in primary samples but decrease less in organoids and are significantly higher in most organoid stages than primary samples." This must be tested (PERMANOVA of salient dims on lineage), and if confirmed, complemented by a lineage-conditional scVI covariate model.
5. **Convergence and immaturity are confounded by construction** and only Control 3 separates them. The honest prior is that some — possibly all — apparent convergence is shared immaturity; the protocol is built to detect and report that outcome as a legitimate result.

Details

PART A — HARD PIECE 1: REAL scArches SURGERY ON RECOVERED RAW UMI COUNTS

A.1 Environment and versions (2026)

- **scvi-tools 1.4.3** (released May 12, 2026; requires Python ⩾3.12, supports 3.12–3.14; "Copyright (c) 2026, Yosef Lab, Weizmann Institute of Science"). Install in a clean env (`conda create -n scvi-env python=3.12`). scArches surgery is built into scvi-tools via `ArchesMixin`; the standalone `scarches` package (`sca.models.SCVI`/`SCANVI`) wraps the same logic and is interchangeable. Prefer native scvi-tools for the SCVI/SCANVI path.
- Supporting: `scanpy` ⩾1.10, `anndata` ⩾0.10, `scib`/`scib-metrics` for integration metrics, `cytotrace2-py` v1.1.0, `POT` (pythonot) ⩾0.9.6, `giotto-tda` ⩾0.6, `scikit-learn`, `skbio` (PERMANOVA).
- GPU: a single NVIDIA A100 40/80GB or H100 is ample. 2026 spot pricing is roughly $1.0–1.4/GPU-hr (A100 80GB; e.g., Thunder Compute lists A100 80GB at $0.78/hr, RunPod A100 PCIe at ~$1.39/hr) and $1.5–3/GPU-hr (H100). Budget envelope below.

A.2 Why NB/ZINB requires integer counts (the mathematical reason for the NaN)

scVI's generative model (Lopez 2018) draws observed counts from a count distribution — by default ZeroInflatedNegativeBinomial, optionally NegativeBinomial or Poisson — parameterized by a decoded mean (a softmax over genes scaled by library size), an inverse-dispersion θ, and (for ZINB) a dropout logit. The NB log-likelihood contains `lgamma(x + θ) - lgamma(x + 1) - lgamma(θ)` and `x·log(μ/(μ+θ))` terms, defined for non-negative integers `x`. For non-integer `x` the model is mis-specified, and — critically — when values are large/continuous (length- or log-normalized) the `μ`/`θ` estimates and their gradients blow up, yielding `inf`/`NaN` that propagate through backprop. During surgery (few free parameters, small query, high learning-rate sensitivity), this is exactly what produced the three divergences. The pilot's 0-epoch fallback simply froze the model so the bad likelihood was never optimized — at the cost of no query adaptation. scvi-tools itself warns: *"Make sure the registered X field in anndata contains unnormalized count data."*

Verification before any training (mandatory gate):

- Integrality: `X = adata.layers['counts']; assert np.allclose(X.data, np.round(X.data))` (sparse-safe on `.data`).
- Non-negativity: `assert X.min() >= 0`.
- Library-size sanity: per-cell sum `np.asarray(X.sum(1)).ravel()` should be plausible UMI totals (hundreds–tens of thousands), not ~1.0 (CPM-rounded) or tiny (length-normalized). Flag medians <200 or >1e5.
- No all-zero genes after subsetting to reference `var_names` (all-zero columns destabilize per-gene dispersion). Drop or, better, ensure HVG set matches reference.

A.3 Recovering genuine counts (the operational bottleneck)

The HNOCA public AnnData is **not documented to carry raw integer UMIs** in a named layer; its tooling references `lognorm` and `scvi_normalized`. Therefore:

1. **First, inspect the actual object**: open the Zenodo `hnoca_cleanedmeta.h5ad` (and the larger "full" object) and enumerate `adata.layers.keys()`, `adata.raw`, and check integrality of each candidate. If a `counts` layer exists and passes A.2, use it directly — skip recovery.
2. **If not, re-derive per-study from primary deposits** listed in HNOCA Supplementary Table 1 (curated via the sfaira framework, per the Methods). For 10x studies this means the GEO/ArrayExpress filtered count matrices (`.mtx`/h5). Concatenate per-study, intersect to the reference `var_names`, and rebuild `adata.layers['counts']`.
3. **Length-normalized layers cannot be round-tripped to ZINB-valid pseudo-counts.** Length normalization divides counts by gene length — appropriate for full-length protocols, *not* 10x 3' (a 3'-tag protocol, so the layer is anomalous and signals a non-count transformation). There is no invertible map from length-normalized values back to integers without the original counts, the exact length vector, and the library factors; multiplying back introduces non-integers and per-gene bias. Do not attempt it. If a study yields *only* normalized layers and the raw deposit is unrecoverable, that study is **counts-unrecoverable** → handle via the branch in A.7.

A.4 Reference-mapping surgery — exact API and configuration

Assuming the reference SCANVI model (`ref_model/`) was trained on the harmonized cross-atlas reference with raw counts:

python

```python
import scvi, scanpy as sc, numpy as np

# 1. Make query var match reference exactly (order + identity)
scvi.model.SCANVI.prepare_query_anndata(adata_query, "ref_model/")
#   -> pads missing genes with zeros, reorders to reference var_names,
#      drops extras. Logs "Found X% reference vars in query data."
#      Require >=80% found; investigate if lower (gene-ID build mismatch).

# 2. Register the counts layer + unknown labels for query
#    (setup is inherited from the reference registry; the query's
#     batch_key categories are appended as NEW codes to the reference)
q = scvi.model.SCANVI.load_query_data(
        adata_query,
        "ref_model/",
        freeze_dropout=True,        # scArches convention
        # defaults retained (these implement the "freeze" surgery):
        # unfrozen=False,
        # freeze_expression=True,
        # freeze_decoder_first_layer=True,
        # freeze_batchnorm_encoder=True,
        # freeze_classifier=True,
)
# fully-unlabeled query:
q._unlabeled_indices = np.arange(adata_query.n_obs)
q._labeled_indices = []

# 3. Train ONLY the new conditional/batch terms
q.train(
    max_epochs=150,
    plan_kwargs=dict(weight_decay=0.0),     # do NOT decay reference weights
    early_stopping=True,
    early_stopping_monitor="elbo_validation",
    early_stopping_patience=15,
    check_val_every_n_epoch=5,
    batch_size=256,
)
adata_query.obsm["X_scANVI"] = q.get_latent_representation()
pred = q.predict()   # transferred labels (SCANVI advantage over SCVI)
```

Why `weight_decay=0.0`. scArches surgery freezes the trained reference weights and adds a new set of conditional (batch) nodes/weights for the query study; only those new weights (and, depending on freeze flags, batchnorm/first-layer terms) are fine-tuned. Weight decay would pull the retained reference weights toward zero, corrupting the learned manifold; the scvi-tools and scArches tutorials therefore standardize `plan_kwargs={"weight_decay": 0.0}` for query training. This is the parameter-efficient "architectural surgery." Python

Freeze flags (defaults shown in the `SCANVI.load_query_data` API): `freeze_expression=True`, `freeze_decoder_first_layer=True`, `freeze_batchnorm_encoder=True`, `freeze_batchnorm_decoder=False`, `freeze_classifier=True`, `freeze_dropout=False` (tutorials set it `True`). **Frozen surgery** (`unfrozen=False`, default) trains only the new query-conditional terms — the conservative, signal-preserving choice and the correct default for NullState. **`unfrozen=True`** retrains the whole network end-to-end — higher batch correction but the highest over-correction risk; use only as a diagnostic comparator, never as the primary. Project name not set

Epochs/early-stopping. scArches/scvi-tools tutorials use up to a few hundred epochs but rely on ELBO early-stopping (`elbo`/`elbo_validation`) for the unsupervised/surgery phase and classifier `accuracy` for the SCANVI semi-supervised phase. Cap `max_epochs=150–200` and let ELBO early-stopping (`patience≈15`) end it; **deliberately limited epochs are themselves an over-correction safeguard** (see A.6).

Likelihood/dispersion for cross-assay query. Use **`gene_likelihood='nb'`** rather than `'zinb'`. Rationale: (i) modern UMI 10x data has limited true zero-inflation, so NB is usually sufficient and is numerically more stable; (ii) ZINB's extra dropout logit is the most fragile term under cross-assay shift and a common NaN source. Use **`dispersion='gene'`** (constant per gene) for robustness; `'gene-batch'` better models assay-specific overdispersion but adds parameters that, with a small/idiosyncratic query batch, can overfit and *increase* over-correction — test only as a sensitivity comparator. The *reference* model's likelihood/dispersion is fixed at reference-training time and inherited by the query; if the reference was trained with ZINB and is unstable on the query, retrain the reference with NB (feasible within budget).

A.5 Over-correction vs under-correction — the central uncertainty and its diagnostics

The danger: real surgery succeeds at batch mixing so well that genuinely off-manifold off-target cells (e.g., mesenchyme) are pulled onto the neural reference manifold and re-labeled as neural — destroying the very signal NullState measures. The opposite failure (under-correction) leaves query cells stranded in their own batch island, inflating ambiguous_novel (the pilot's 37.7%).

Diagnostic battery (run pre- vs post-surgery, and frozen vs unfrozen):

Quantity	Tool	Over-correction signature	Under-correction signature	Pass target
Batch mixing query↔ref	graph iLISI, kBET acceptance	iLISI high everywhere incl. off-target	iLISI ≈1 (no mixing)	iLISI improved vs 0-epoch; kBET acceptance materially up on *on-target* cells
Bio conservation	graph cLISI, cell-type ASW	cLISI degrades (types blur)	cLISI fine but isolated query island	cLISI maintained near reference-internal level
Off-target retention	fraction in ambiguous_novel + true_offtarget	true_offtarget collapses toward 0	ambiguous_novel stuck ~37.7%	true_offtarget rises from 17.8% AND biologically coherent
Held-out query reconstruction	`q.get_reconstruction_error()` on held-out query	not improved despite mixing	high	decreases vs 0-epoch projection
Marker coherence (decisive)	mean expression of COL1A1, COL3A1, DCN, LUM, PDGFRA, PDGFRB in cells that *moved* ambiguous_novel→true_offtarget	movers NOT mesenchyme-high (random reassignment)	n/a	movers are mesenchyme-marker-high (coherent)

Pass/fail rule for surgery acceptance: Accept iff (1) iLISI/kBET on *on-target* neural cells improves meaningfully over the 0-epoch baseline; (2) cLISI / cell-type ASW does **not** degrade by more than a pre-registered margin (≤5% relative); (3) held-out query reconstruction error decreases; and (4) cells reassigned into true_offtarget are mesenchyme-marker-coherent (mean COL1A1/COL3A1/DCN/LUM/PDGFRA/PDGFRB z-score clearly elevated vs ambiguous_novel cells that did not move, e.g., Cohen's d > 0.8). If batch mixing improves but cLISI degrades and movers are marker-incoherent → **over-correction**: reduce `max_epochs`, keep frozen surgery, drop `gene-batch` dispersion, and/or raise latent dimensionality so off-target variance has somewhere to live.

Use scIB/scib-metrics for the composite (kBET, graph iLISI/cLISI, graph connectivity, PCR, cell-type ASW, isolated-label scores). Report the batch-correction vs bio-conservation trade-off explicitly (scIB's 40:60 weighting convention), but **do not optimize the composite** — for NullState, bio-conservation of the off-target population outweighs maximal mixing, so weight cLISI/marker-coherence above iLISI in the decision.

A.6 Tuning to retain query-specific structure

- Prefer **frozen** surgery + **few epochs** — the principled lever to retain off-target structure.
- Keep `n_latent` ⩾ the reference's (⩾10; consider 20–30 if the reference used few dims) so novel off-target axes are representable.
- Compare latent neighborhoods pre/post: for each off-target cell, the fraction of k-NN that are reference cells should rise modestly, not saturate to ~1.0. Saturation = over-correction.

A.7 Failure-mode decision tree (Hard Piece 1)

1. **Still diverges to NaN with real counts** → (a) re-confirm integrality/no all-zero genes; (b) switch `zinb`→`nb`; (c) lower LR via `plan_kwargs=dict(lr=1e-4, weight_decay=0.0)`; (d) enable `reduce_lr_on_plateau` and gradient clipping (`gradient_clip_val≈1.0` via `trainer_kwargs`); (e) Poisson fallback (`gene_likelihood='poisson'`) — most numerically robust, drops dispersion; (f) reduce `batch_size` to 128.
2. **Query has batches not in reference** → expected; `load_query_data` appends new batch codes and surgery trains exactly those. Set the batch covariate at the granularity the reference used (HNOCA used a dataset/sample/tech-sample concatenation; for a query, treat each study/dataset as one batch). If a query batch is tiny (<~200 cells) it will be poorly corrected — merge or down-weight.
3. **Only a subset of studies have recoverable counts** → run surgery on the counts-recoverable subset only; treat counts-unrecoverable studies as a separate, clearly-labeled cohort analyzed by reference projection only (0-epoch), never pooled into convergence statistics. Report n recovered vs total.
4. **Gene-order/var mismatch** (`Found <80% reference vars`) → almost always a gene-ID build mismatch (Ensembl version, symbol vs ID). Harmonize IDs before `prepare_query_anndata`; never proceed on a low-overlap mapping.
5. **HVG mismatch** → the query must be subset to the reference's exact training genes (handled by `prepare_query_anndata`), not re-HVG'd.

PART B — HARD PIECE 2: contrastiveVI DISH-STRIPPING + MATURATION-MATCHED CONTROL 3

B.1 contrastiveVI theory and exact configuration

`scvi.external.ContrastiveVI` (Weinberger E, Lin C, Lee S-I, *"Isolating salient variations of interest in single-cell data with contrastiveVI,"* Nat Methods 20:1336–1345, 2023, doi:10.1038/s41592-023-01955-3; corresponding author Su-In Lee, Univ. Washington) decomposes variation into a **background latent** (shared by background and target) and a **salient latent** (active only in target; fixed to a zero vector for background). A single shared decoder reconstructs all cells, which pressures background variables to capture common structure and salient variables to capture target-specific variation. It uses a ZINB/NB count likelihood → **requires raw counts,** tying directly to Hard Piece 1. Project name not set

NullState design (the key conceptual mapping):

- **background = primary on-target cells** (in vivo; no dish signature)
- **target = organoid on-target cells** (carry the dish signature)
- **salient latent → the dish/in-vitro axis** (enriched in organoid vs primary)
- **background/shared latent → lineage biology free of dish** (the space in which convergence is later measured)

python

```python
import scvi, numpy as np
scvi.external.ContrastiveVI.setup_anndata(adata, layer="counts")  # RAW counts
m = scvi.external.ContrastiveVI(
        adata,
        n_background_latent=10,
        n_salient_latent=10,
        use_observed_lib_size=True,   # observed library size as scaling factor
        wasserstein_penalty=0.0,      # raise (e.g., 1-10) if shared leaks into sali
)
m.train(
    background_indices=np.where(adata.obs["origin"]=="primary")[0],
    target_indices=np.where(adata.obs["origin"]=="organoid")[0],
    max_epochs=500, early_stopping=True,    # ELBO early-stopping
)
sal = m.get_latent_representation(adata, representation_kind="salient")
shr = m.get_latent_representation(adata, representation_kind="background")
```

The `wasserstein_penalty` term (default 0) is documented as the "weight of the Wasserstein distance loss that further discourages background shared variations from leaking into the salient latent space" — the lever to use when disentanglement is imperfect.

B.2 Proving disentanglement (quantitative, pre-registered thresholds)

- **(a) Origin predictable from salient:** logistic-regression / random-forest classifier (organoid vs primary) on the salient latent, stratified CV. **Pass: AUROC > 0.85.**
- **(b) Origin NOT predictable from shared:** same classifier on the background/shared latent. **Pass: AUROC ≈ 0.5–0.6** (≤0.65 hard ceiling). If shared AUROC is high → leakage → raise `wasserstein_penalty`, reduce `n_salient_latent`, or retrain.
- **(c) Salient tracks stress, shared does not:** score every cell for **HALLMARK_GLYCOLYSIS, HALLMARK_HYPOXIA, HALLMARK_UNFOLDED_PROTEIN_RESPONSE** (MSigDB Hallmark; via `scanpy.tl.score_genes` on log-norm, or AUCell/decoupleR for rank-based robustness). Correlate each score with salient and shared dimensions (canonical correlation or per-dim Spearman). **Pass: salient–stress |r| substantially higher than shared–stress** (e.g., salient max |r| > 0.4 while shared max |r| < 0.2). Add organoid-specific markers for a domain check, anchored on Bhaduri et al. 2020 (which reported "a dramatic increase in the expression of glycolysis gene PGK1 and the ER stress gene, GORASP2"): glycolysis PGK1; ER/UPR ARCN1, GORASP2, DDIT3/CHOP, XBP1, DDIT4, P4HB; hypoxia SLC2A1/GLUT1, PDK1.

B.3 Single vs conditional salient subspace

The organoid-stress literature (Bhaduri 2020; the "stressed-state" subpopulation work, bioRxiv 2022; Gruffi) shows glycolytic/hypoxic shifts are **pan-organoid** (diffusion-limited hypoxia in the organoid core, shared across lineages) whereas **ER/UPR stress is lineage-modulated** (scales with secretory/protein-folding demand). Bhaduri et al. 2020 explicitly found that "ER stress and glycolysis networks decrease over time in primary samples but decrease less in organoids and are significantly higher in most organoid stages than primary samples." A single salient subspace may therefore under-fit a partly lineage-dependent dish signature.

Test: in the salient latent, run **PERMANOVA** (`skbio.stats.distance.permanova` on a Euclidean distance matrix) or **distance-based redundancy analysis (db-RDA)** with lineage as the factor; quantify the fraction of salient variance explained by lineage. Complement with per-lineage salient centroids (offsets): large systematic per-lineage offsets ⇒ lineage modulation. **Decision threshold:** if lineage explains a substantial, significant share of salient variance (e.g., R^2 > ~0.15, p<0.01 by permutation), the single salient axis is inadequate.

If lineage-modulated, complement contrastiveVI with **a lineage-conditional model**:

- **scVI with the dish/condition as a covariate to regress out** (`setup_anndata(..., categorical_covariate_keys=["origin"])` or as batch), giving a condition-adjusted latent; or
- **the project's prior fallback of a conditional `z_iv | cell-type`** — fit the dish vector separately within each broad lineage and subtract per-lineage; or
- a **cVAE variant / ContrastiveVI+** (richer per-condition salient prior).

Honest comparison. contrastiveVI is the most principled because it *explicitly* separates shared vs target-specific variation and is purpose-built for the target-vs-background ("contrastive analysis") setting; its weakness is identifiability — leakage between spaces is documented, and a single salient prior can shrink small or heterogeneous effects (the explicit motivation for ContrastiveVI+). scVI-covariate regression is simpler, more stable, and trivially lineage-conditional, but it does not *isolate* the dish axis (it removes condition variance without modeling it), so you lose the ability to validate the dish axis directly via B.2. The crude mean-difference dish-vector used in the pilot (mean cosine 0.906 across cell types) is the weakest: it assumes one global linear dish direction and cannot represent lineage modulation — its high cross-cell-type cosine is *consistent with* but does not *prove* a shared dish axis, and it has no maturation control. **Recommendation: contrastiveVI as primary, scVI-covariate as the lineage-conditional cross-check; report convergence only where both agree.**

B.4 MATURATION-MATCHED CONTROL 3 (the deepest methodological point)

The confound. Organoid off-target cells are developmentally immature; immature cells of *different* lineages resemble each other simply by being immature. So cross-lineage "convergence" could be a trivial consequence of shared immaturity, not a genuine shared default/attractor state. Control 3 separates these.

(a) Lineage-independent, dish-robust maturation scoring — CytoTRACE2 (primary). CytoTRACE2 (Kang M, Gulati GS, Brown EL, Qi Z et al., Nat Methods 22:2258–2263, 2025, doi:10.1038/s41592-025-02857-2; `cytotrace2-py` v1.1.0) is an interpretable deep-learning method predicting **absolute** developmental potency on a fixed [0,1] scale (0 = differentiated, 1 = totipotent) — not a dataset-relative ranking. The v1.1.0 package "relies on an ensemble of … per-dataset trained models" (trained over 17 datasets spanning 18 diverse human/mouse tissues; the retrained framework expands the ensemble to 19 models). It is explicitly designed to **suppress batch/platform variation** (competing rank-based + log2 representations, training-set diversity, adaptive KNN smoothing) — exactly why it is the right maturation axis for a dish-confounded comparison.

python

```python
from cytotrace2_py.cytotrace2_py import cytotrace2
# input: genes x cells .txt (raw OR CPM/TPM; NOT log-transformed)
results = cytotrace2(
    "expr_genes_by_cells.txt",
    annotation_path="annot.txt",   # optional, for plots
    species="human",               # default is "mouse" - MUST set
    batch_size=10000,
    smooth_batch_size=1000,
    seed=14,
)
# -> CytoTRACE2_Score in [0,1], CytoTRACE2_Potency category, CytoTRACE2_Relative
```

Use the continuous `CytoTRACE2_Score` as the maturation coordinate. (Manuscript-reproduction settings are `batch_size=100000, smooth_batch_size=10000`; the smaller defaults are fine and cheaper. The third-party `omicverse.single.cytotrace2` wrapper accepts AnnData directly with `max_pcs=200` if you prefer not to write `.txt`.) GitHub GitHub

Why RNA-velocity latent time is a desirable cross-check but is BLOCKED here. scVelo's dynamical model (Bergen et al., Nat Biotechnol 38:1408–1414, 2020) recovers a gene-shared **latent time** from spliced/unspliced mRNA — an orthogonal, dynamics-grounded maturation axis independent of CytoTRACE2's expression-rank model. But it **requires spliced/unspliced count layers**, which are *not posted* for these studies; re-quantifying them from BAM/FASTQ (velocyto/kallisto|bustools) is out of scope on the budget (terabytes of raw reads, large compute). Latent time is therefore **unavailable** — document this as a known limitation. *If* it were available, the disciplined use would be to restrict Control 3 to **concordant-rank cells** (cells whose CytoTRACE2 and latent-time maturation ranks agree) to avoid maturation-axis artifacts.

(b) Maturation-matched primary baseline. Build a primary comparison set whose CytoTRACE2 histogram matches the organoid off-target/convergent population, jointly with broad lineage:

1. Bin CytoTRACE2 score into K bins (e.g., 10 equal-width or quantile bins).
2. For the organoid convergent-state population, compute its joint (maturation-bin × broad-lineage) histogram.
3. **Stratified resampling** of primary cells to match that joint histogram per (bin, lineage). Match on **both** maturation and lineage so you are not accidentally matching maturation by swapping lineage composition.
4. **Power consequence:** matching discards primary cells in over-represented bins, shrinking N and inflating the self-distance floor. Fix the minimum stable N empirically (rarefaction, below). If a (bin,lineage) cell is empty on the primary side, that organoid stratum is **unmatchable** → report separately, do not extrapolate.

(c) The discriminating test — dimension-robust optimal transport in the dish-stripped shared latent. Compute distances in the **contrastiveVI shared/background latent** (dish removed), **never raw high-dim Wasserstein** (empirical W convergence is O(n^−1/d) — the curse of dimensionality).

- **Primary statistic — sliced-Wasserstein:** `ot.sliced_wasserstein_distance(X, Y, n_projections=200)` (POT) — averages 1-D Wasserstein over ⩾200 random projections; sample complexity O(n^−1/2), independent of dimension.
- **Cross-check — debiased Sinkhorn divergence:** `ot.bregman.empirical_sinkhorn_divergence(X, Y, reg)` — debiased so S(μ,μ)=0; report alongside SWD for robustness to the projection choice.

The convergence logic (pre-registered): convergence is real **only if**

1. d(organoid-convergent-state, **maturation-matched** primary) is significantly **below** a cross-lineage divergence scale (the typical distance between distinct primary lineages), **AND**
2. it is **below** d(organoid-convergent-state, **unmatched** primary) — matching for maturation does *not* erase the closeness, **AND**
3. it is **above** the matched-N self-distance floor (otherwise indistinguishable from sampling noise).

Null and error control:

- **Matched-N self-distance floor:** repeated split-half of the convergent population, SWD between halves at matched N → the noise floor for that N.
- **Permutation null (⩾1000 perms):** shuffle origin/lineage labels **within maturation bins** (preserving the maturation distribution under the null) and recompute SWD; p = fraction of permuted distances ⩽ observed. Within-bin shuffling is essential — it tests "closer than expected *given equal maturity*," which is exactly the immaturity confound.
- **FDR** (Benjamini–Hochberg) across lineage-pair / stratum tests.
- **Minimum stable N by rarefaction:** subsample ladder (100, 200, 400, 800, 1600); plot split-half self-distance vs N; use the N at which the curve saturates as the minimum for all reported comparisons.
- **Power:** no closed form for a permutation test on SWD → estimate by **simulation**: inject known shifts of varying magnitude into synthetic populations at the chosen N and measure detection rate.

(d) Topology as confirmatory (not primary). Compute **H0 persistent homology** with `giotto-tda` `VietorisRipsPersistence(homology_dimensions=[0], collapse_edges=True)` on the shared-latent point clouds, with **Fasy-style bootstrap confidence bands** (subsample/bootstrap the diagram, take the (1−α) quantile of bottleneck distances to define a significance band — the same logic as the TDA-package `bootstrapDiagram`). H0 measures connected-component structure (do organoid-convergent and matched-primary merge into one component at low filtration?). **H1 (loops) is exploratory only** — interpret cautiously, do not base conclusions on it.

(e) The decisive failure-mode branch — call discipline.

- If **matched distance ≈ self-distance floor** (not significant vs the within-bin permutation null): convergence is **fully explained by shared immaturity**. The "default state / attractor" claim is **NOT supported**. Reframe — still a legitimate, publishable result — as a **"shared-immature / developmentally-arrested state."**
- If matched distance is significantly below the cross-lineage scale and above the floor, **and** survives leave-one-study-out (below): convergence beyond immaturity is supported.
- **Language discipline (mandatory):** do **not** write "attractor" without dynamical evidence (which RNA velocity would have provided and which is blocked here). Permitted terms: **"convergent state," "low-dispersion configuration," "shared transcriptomic state."** "Attractor" implies a dynamical basin and is unsupported by static snapshots.

B.5 Interaction of the two pieces, sequencing, and the study-concentration guard

- **Hard Piece 1 gates Hard Piece 2.** contrastiveVI needs raw counts (same recovery as A.3). More subtly, a **soft-edged off-target population from bad surgery poisons the convergence analysis**: if over-correction has absorbed real mesenchyme into the neural manifold, the "convergent state" is an artifact of mislabeling; if under-correction leaves a smeared ambiguous_novel blob, the population fed to contrastiveVI is contaminated. **Sequence:** (1) recover counts → (2) real surgery + accept/reject by A.5 → (3) define a clean off-target/convergent population → (4) contrastiveVI dish-stripping → (5) disentanglement validation B.2/B.3 → (6) CytoTRACE2 + Control 3.
- **Gate-C concentration caveat — wrap BOTH pieces in leave-one-study-out (LOSO).** The off-target mesenchyme is heavily concentrated in a small number of studies (your pilot attributes ~61.7% to one Treutlein-lab/Fleck–He morphogen-screen dataset). *Caveat on this figure:* the published core HNOCA actually reports non-neuroectodermal lineages (mesenchyme, endothelium, immune) as largely **absent/under-represented** — "This analysis confirmed the absence of erythrocytes, immune cells and vascular endothelial cells in the HNOCA" — with most off-target/non-neural content arising from morphogen-screen datasets (Amin et al. 2023, mapped in HNOCA Fig. 4; and the separate Sanchís-Calleja/Fleck/Treutlein screen, Nat Methods 2025). Verify the 61.7% against your own assembled cohort rather than the core atlas. Regardless of the exact share, a dominant study must not drive either result: ResearchGate nih
  - **Surgery re-classification:** re-run the accept/reject diagnostics (A.5) with the dominant study held out; true_offtarget recovery and marker-coherence must hold without it.
  - **Convergence statistics:** recompute SWD/Sinkhorn, permutation null, and the Control-3 call with each study held out in turn (and/or per-study down-weighting in the resampling so no study exceeds a capped weight). Report convergence **only if it survives LOSO**.

Recommendations (staged, with go/no-go thresholds)

1. **Stage 0 — Counts audit (first; ~1 day, ~$0).** Open the HNOCA Zenodo objects, enumerate layers/`.raw`, test integrality (A.2). **Go** if a valid counts layer exists. **No-go → Stage 0b:** re-derive per-study counts from Supplementary-Table-1 accessions via sfaira. Record which studies are counts-recoverable; this set defines everything downstream.
2. **Stage 1 — Reference likelihood check.** Confirm the reference SCANVI model uses `nb` (retrain with `nb` if it is `zinb` and unstable). ~1–2 GPU-hr.
3. **Stage 2 — Real surgery** (A.4): frozen surgery, `weight_decay=0.0`, `nb`, capped epochs, ELBO early-stopping. Run the A.5 battery pre/post and frozen-vs-unfrozen. **Accept** only on the 4-part rule (iLISI/kBET up on-target; cLISI preserved; held-out reconstruction down; movers mesenchyme-coherent). ~2–4 GPU-hr.
4. **Stage 3 — Define the clean off-target/convergent population** from accepted surgery. Recompute the ambiguous_novel vs true_offtarget split; report new fractions vs the pilot's 37.7%/17.8%.
5. **Stage 4 — contrastiveVI dish-stripping** (B.1) on raw counts; **prove disentanglement** (B.2). **Go** only if salient-origin AUROC>0.85, shared-origin AUROC≈0.5–0.6, and salient (not shared) tracks the three Hallmark stress sets. Otherwise raise `wasserstein_penalty`/reduce `n_salient_latent` and retrain. Test single-vs-conditional salient (B.3); add scVI-covariate cross-check if lineage-modulated. ~2–4 GPU-hr.
6. **Stage 5 — CytoTRACE2 + Control 3** (B.4): score maturation, build the maturation×lineage-matched primary baseline, run SWD (n_projections⩾200) + Sinkhorn cross-check in the shared latent, permutation null (⩾1000, within-bin shuffles), rarefaction for min N, simulation power, H0 topology confirmatory. Mostly CPU; budget wall-clock, not GPU.
7. **Stage 6 — LOSO everywhere** (B.5). Report convergence only if it survives leave-one-study-out and the matched distance is significantly below cross-lineage scale and above the self-distance floor. If matched ≈ floor → publish the **shared-immature/arrested-state** reframing with the mandated language discipline.

Thresholds that change the plan: reference-var overlap <80% → stop and fix gene IDs. cLISI degradation >5% with marker-incoherent movers → over-correction, reduce epochs/keep frozen. Shared-latent origin AUROC >0.65 → leakage, do not trust dish-stripping. Lineage R^2>~0.15 in salient → switch to lineage-conditional dish model. Matched SWD within the permutation null of the floor → reframe to immaturity.

Budget: all GPU training (reference retrain + surgery + contrastiveVI) is well under ~15 GPU-hr total; at A100 spot ≈$1.0–1.4/hr that is ~$15–25, leaving the bulk of the $150–200 for storage, egress, and CPU permutation/rarefaction time. The project is comfortably feasible on a single GPU within budget; the true bottleneck is the **counts-recovery labor**, not compute.

Caveats

- **Counts provenance is unconfirmed.** I could not verify from public docs that the HNOCA release carries raw integer UMIs in a named layer; documented layers are `lognorm`/ `scvi_normalized`. Stage 0 must resolve this empirically. HNOCA was integrated with **scPoli** (NB loss, raw counts), not the SCVI/SCANVI path your reference uses — confirm which model your reference actually is, since scPoli surgery uses `sca.models.scPoli` rather than `SCANVI.load_query_data`.
- **The 61.7%/single-study mesenchyme figure is internally inconsistent with the published core atlas**, which reports non-neural lineages as largely absent; off-target mesenchyme is associated chiefly with morphogen-screen datasets (Amin 2023; Sanchís-Calleja/Fleck/Treutlein 2025). Verify the concentration in your own assembled cohort. Either way, LOSO is mandatory. ResearchGate
- **RNA-velocity latent time is unavailable** (no spliced/unspliced layers; re-quantification out of scope), so the dynamical cross-check on maturation and any literal "attractor" claim are out of reach. Use static-snapshot language only.
- **contrastiveVI identifiability is not guaranteed**; leakage between salient and shared spaces is documented in the comparative-deep-generative-model literature. The B.2 validation is not optional — it is the only thing standing between a real dish-stripping and a confounded one.
- **scvi-tools surgery API specifics** (freeze-flag defaults, `plan_kwargs` location) are stable across 1.x but exact signatures can shift between minor versions; pin the version (1.4.3) and re-read the installed `SCANVI.load_query_data` docstring before running. The `weight_decay=0.0` and `freeze_dropout=True` conventions are consistent across documented versions.
- **`full_model` for cytotrace2-py** is confirmed only as a CLI flag (`--full-model`/`-fm`); confirm whether it is exposed as a Python keyword in the installed v1.1.0 before relying on it. Note also that v1.1.0 requires input to be raw or CPM/TPM (not log-transformed).
- All pass/fail thresholds here are pre-registration-grade defaults; fix them before looking at outcomes to avoid the garden-of-forking-paths.